

How to interpret GLM coefficients?

Asked 7 years, 11 months ago Modified 7 years, 11 months ago Viewed 14k times

I am reproducing the results from COMPAS analysis done by [propublica](#) and I needed some help understanding how they handled interpretation of GLM coefficients. Score_factor is a variable indicating risk of recidivism and its regressed against variables like race, gender etc. The model is given below.

Call:

```
glm(formula = score_factor ~ gender_factor + age_factor + race_factor +
    priors_count + crime_factor + two_year_recid, family = "binomial",
    data = df)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.52554	0.07851	-19.430	< 2e-16 ***
gender_factorFemale	0.22127	0.07951	2.783	0.005388 **
age_factorGreater than 45	-1.35563	0.09908	-13.682	< 2e-16 ***
age_factorLess than 25	1.30839	0.07593	17.232	< 2e-16 ***
race_factorAfrican-American	0.47721	0.06935	6.881	5.93e-12 ***
race_factorAsian	-0.25441	0.47821	-0.532	0.594717
race_factorHispanic	-0.42839	0.12813	-3.344	0.000827 ***
race_factorNative American	1.39421	0.76612	1.820	0.068784 .
race_factorOther	-0.82635	0.16208	-5.098	3.43e-07 ***
priors_count	0.26895	0.01110	24.221	< 2e-16 ***
crime_factorM	-0.31124	0.06655	-4.677	2.91e-06 ***
two_year_recid	0.68586	0.06402	10.713	< 2e-16 ***

From the model above, they concluded that "Black defendants are 45% more likely than white defendants to receive a higher score correcting for the seriousness of their crime, previous arrests, and future criminal behavior." (Note race_factorWhite is part of intercept) based on following calculation:

```
control <- exp(-1.52554) / (1 + exp(-1.52554))
exp(0.47721) / (1 - control + (control * exp(0.47721)))
[1] 1.452841
```

I am aware that we can get change in odds ratio of score_factor between white and black defendants by doing exp(0.47721). But I am not sure what the calculation with control and intercept is doing. Can someone please explain?

regression generalized-linear-model

Share Cite Edit Follow Flag

asked Sep 2, 2016 at 16:28

Rajesh

93 1 1 8

1 Answer

Sorted by: Highest score (default)

The original author of the analysis was kind enough to respond and clear this up for me. The calculation converts odds ratio (which are highly unintuitive) to relative risk (which is easier to understand). More details are provided [here](#).

The GLM coefficients only show the multiplicative change in odds ratio. so if p1 is the risk of getting a high score for black defendants and p0 is the risk of getting a high score for white defendants, then exp(0.47721) shows (p1/(1-p1))/(p0/(1-p0)). Unfortunately, this is not particularly easy to intuit. Relative risk (p1/p0) is a much simpler concept to understand and odds ratio can be converted to relative risk using the formula: Relative risk=odds ratio/(1-p0+(p0×odds ratio)). The calculation in the question accomplishes that.

Share Cite Edit Follow Flag

edited Sep 8, 2016 at 19:54

answered Sep 3, 2016 at 17:18

Rajesh

93 1 1 8

It's great that you found an answer! However, leaving all the details behind a link means that your answer becomes useless once [link rot](#) sets in. It will be very helpful to future readers if you could add a short summary of what the link provides. Since it's a link to an academic paper, you might as well go ahead and provide a full academic citation. That Will go along way towards "future proofing" your answer. – shadowtalker Sep 3, 2016 at 17:43

The GLM coefficients only show the multiplicative change in odds ratio. so if p1 is the risk of getting a high score for black defendants and p0 is the risk of getting a high score for white defendants, then exp(0.47721) shows (p1/(1-p1))/(p0/(1-p0)). Unfortunately, this is not particularly easy to intuit. Relative risk (p1/p0) is a much simpler concept to understand and odds ratio can be converted to relative risk using the formula: Relative risk=odds ratio/(1-p0+(p0×odds ratio)). The calculation in the question accomplishes that. – Rajesh Sep 7, 2016 at 17:10

You should edit that into your main post (comments aren't searchable by users). You can use the "edit" button at the bottom left – shadowtalker Sep 7, 2016 at 19:18

@Rajesh In most cases you can estimate the relative risk directly with relative risk regression. Relative risk regression is a glm with binomial variance formula (var = mean * (1-mean)), but uses a log link. Attempt the analysis again with family=binomial(link=Log) and you should get comparable results. I cannot advocate enough transforming ORs to RRs when possible. – AdamO Sep 8, 2016 at 19:58